

Due on monday, October 8th, before 5pm

The questions indicated as [**SELF**] are for self-study and additional exercise:
no need to submit your work on these questions.

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1. Consider a two-state system, so that the Hilbert space is two-dimensional, i.e., the Hamiltonian is a 2×2 matrix and the wavefunction is a vector with two complex components. Let's denote the wavefunction as

$$\psi(t) = \begin{pmatrix} u(t) \\ v(t) \end{pmatrix}$$

We will consider a relatively simple (diagonal) Hamiltonian:

$$H = \begin{pmatrix} \epsilon_1 & 0 \\ 0 & \epsilon_2 \end{pmatrix}$$

- (a) [**6 pts.**] Using the time-dependent Schroedinger equation, write down differential equations for the functions $u(t)$ and $v(t)$.
- (b) [**6 pts.**] Solve the differential equations, so that $u(t)$ and $v(t)$ are expressed in terms of the initial values $u(0)$ and $v(0)$.
- (c) [**5 pts.**] Show that the norm of the wavefunction, $|u(t)|^2 + |v(t)|^2$, is independent of time.
- (d) [**3 pts.**] Provide an expression for $\psi(t)$, in the case that the initial state is

$$\psi(0) = \begin{pmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{pmatrix}$$

2. Electromagnetic waves as particles:

- (a) Visible light has a wavelength in the range of 400-700 nm.
 - i. [**3 pts.**] What is the frequency range of visible light?
 - ii. [**4 pts.**] What is the energy range of a photon of visible light?
- (b) [**4 pts.**] A kitchen microwave operates at frequency 2.5 GHz and at power 750 Watts. How many photons does the microwave emit per second?

- (c) **[SELF]** How many microwave photons [from the previous question] are required to heat up a 0.2-liter glass of water by 10°C ? The heat capacity of water is roughly $4185 \text{ J}/(\text{kg}\cdot\text{K})$, and a liter of water weighs roughly 1kg.
- (d) **[SELF]** A low-power laser (10^{-3} Watts) gives out light of wavelength 633 nm. How many photons does it emit per second?
3. Consider the wavefunction of a single particle confined to the positive half of a one dimensional line, i.e., the wavefunction vanishes in the region $x < 0$. The wavefunction is known to be
- $$\psi(x) = \begin{cases} Ne^{-x/\lambda} & \text{for } x > 0 \\ 0 & \text{for } x < 0 \end{cases}$$
- (a) **[7 pts.]** Calculate the constant N so that the wavefunction is normalized. Keep your answer general: do not assume N to be real. (In case normalization has not yet been covered in class: please look it up. E.g., you could try googling “wavefunction normalization”.)
- (b) **[9 pts.]** Calculate the probability of finding the particle inside the region $x \in [\lambda, 2\lambda]$. Reminder: $|\psi(x)|^2$ is the ‘probability density’.
- (c) **[2+1 pts.]** Sketch a plot of the probability density, $|\psi(x)|^2$, as a function of position x . Show the region on this plot whose area represents the probability you have calculated above. (Hand-drawn sketch, please, no computer printouts.)
- (d) **[SELF]** The given wavefunction is not acceptable as a stationary solution of the time-independent SE. Why?
4. **[SELF]** We need to learn “bra-ket” notation. Please read through Nash’s notes, sections 1 to 4. It will be assumed that you have learned linear algebra previously; it won’t be repeated in class.